

Prabhat Zade

- +917020882015
- prabhatzade45@gmail.com
- Hyderabad, Telangana
- [LinkedIn](#)

Profile Summary

- Results-oriented **Data Science professional** with 2+ years of experience driving innovation through **Machine Learning, Natural Language Processing (NLP), and AI technologies**. Currently serving as a Project Engineer at the Centre for Development of Advanced Computing (C-DAC R&D), Hyderabad, leading **end-to-end AI/ML project development and implementation**.
- Proven expertise in **designing, fine-tuning, and deploying transformer-based models** and building scalable **ML pipelines with practical, result-driven strategies**.
- Hands-on experience in model optimization, deployment of AI applications, and integrating **NLP techniques like tokenization, Named Entity Recognition (NER), and Part-of-Speech (POS) tagging**.
- Strong background in data engineering, cloud infrastructure (**AWS**), and automating **CI/CD** workflows for seamless **MLOps**.
- Automated data processing, embedded text with Hugging Face, stored vectors in **FAISS, ChromaDB**, and delivered user-personalized recommendations via a **Retrieval-Augmented Generation** pipeline integrating LangChain and a Groq LLM. Emphasized secure credential management, modular design, and robust logging

Skills

Python | Machine Learning | Deep Learning | CNN | Natural language processing | Langchain | Gen AI | OpenCV | RAG | SQL | Big Data | Hadoop | Spark | Tableau | Predictive modeling | STATISTICS | Unix Shell Scripting | GitHub | Problem-Solving | Team Collaboration | Communication

Work Experience

Project Engineer | Centre for Development of Advanced Computing (C-DAC R&D), Hyderabad |
December 2022 – Present

Key Contributions:

- Created an **end-to-end text summarization application** using **Hugging Face's Transformer model library**, fine-tuning a pre-trained model with the Sentence Transformers library, **leading to a 15% accuracy improvement**.
- Developed a **comprehensive training and prediction pipeline** that managed data ingestion, validation, transformation, model training, and evaluation, **reducing processing time by 30 minutes**; integrated FastAPI to optimize the application's usability and performance.
- implemented **prompt engineering** strategies, **Retrieval-Augmented Generation (RAG) pipeline**, and integrated a large language model (Groq LLM) for context-aware, user-personalized recommendations.
- Deployed **the application on AWS and worked with the team to streamline CI/CD processes using GitHub Actions**, ensuring smooth updates and efficient maintenance.
- Utilized advanced **NLP techniques, including Tokenization, TF-IDF, word embeddings, text classification, Named Entity Recognition (NER), and Part-of-Speech (POS) tagging**, which boosted model performance.
- Organizing, **analyzing, synthesizing and/ or summarizing data using appropriate methodologies**; gathering user requirements and translating these into technical and system specifications.
- Developing **intricate algorithms based on deep-dive statistical analysis and predictive data modeling** to deepen relationships, strengthen longevity and personalize interactions with customers.
- Mapping **business requirements by undertaking quick gap analysis**; understanding business process and analyzing the workflow to design solutions (as-is, to-be and why-why analysis and freezing).
- Performed **data exploration and visualization with tools like Excel and Tableau**, generating actionable insights that improved decision-making processes.

Associate Data Engineer | phData Solutions Pvt. Ltd. | Jun 2022 – Jul 2022

Key Contributions:

- Gained hands-on experience with **Python, SQL, and Snowflake**, developing a solid foundation in data engineering and analytics through designing, building, and optimizing data pipelines and workflows.
- **Designed, developed, tested, and deployed** end-to-end data pipelines, **acquiring hands-on experience** in all stages of data engineering processes.
- **Implemented Agile methodologies** and utilized tools like **JIRA** and **Scrum frameworks** to ensure efficient project management and seamless team collaboration.
- **Enhanced data quality and operational efficiency** by adopting industry best practices for **data management, optimization**, and performance improvement.
- **Actively contributed** to team discussions, showcasing **collaborative problem-solving skills** to effectively address technical challenges.

Education

P.G. Diploma in Big Data Analytics (PG-DBDA) | Centre for Development of Advanced Computing (C-DAC), Bangalore | Sept '21 - Apr '22 | Score: 67%

Bachelors in Engineering (Computer Science and Engineering) | Sant Gadge Baba Amravati University, Amravati, Maharashtra | Aug '16 - Aug '21 | Score: 92.74%

Interests

New Technology Exploration | Gym | Traveling

Languages

English, Hindi and Marathi.